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Analysis of red blood cell partitioning at bifurcations in simulated microvascular networks

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Partitioning of red blood cells (RBCs) at vascular bifurcations has been studied over many decades using *in vivo*, *in vitro*, and theoretical models. These studies have shown that RBCs usually do not distribute to the daughter vessels with the same proportion as the blood flow. Such disproportionality occurs, whereby the cell distribution fractions are either higher or lower than the flow fractions and have been referred to as classical partitioning and reverse partitioning, respectively. The current work presents a study of RBC partitioning based on, for the first time, a direct numerical simulation (DNS) of a flowing cell suspension through modeled vascular networks that are comprised of multiple bifurcations and have topological similarity to microvasculature *in vivo*. The flow of deformable RBCs at physiological hematocrits is considered through the networks, and the 3D dynamics of each individual cell are accurately resolved. The focus is on the detailed analysis of the partitioning, based on the DNS data, as it develops naturally in successive bifurcations, and the underlying mechanisms. We find that while the time-averaged partitioning at a bifurcation manifests in one of two ways, namely, the classical or reverse partitioning, the time-dependent behavior can cycle between these two types. We identify and analyze four different cellular-scale mechanisms underlying the time-dependent partitioning. These mechanisms arise, in general, either due to an asymmetry in the RBC distribution in the feeding vessels caused by the events at an upstream bifurcation or due to a temporary increase in cell concentration near capillary bifurcations. Using the DNS results, we show that a positive skewness in the hematocrit profile in the feeding vessel is associated with the classical partitioning, while a negative skewness is associated with the reverse one. We then present a detailed analysis of the two components of disproportionate partitioning as identified in prior studies, namely, plasma skimming and cell screening. The plasma skimming component is shown to under-predict the disproportionality, leaving the cell screening component to make up for the difference. The crossing of the separation surface by the cells is observed to be a dominant mechanism underlying the cell screening, which is shown to mitigate extreme heterogeneity in RBC distribution across the networks. *Published by AIP Publishing.* <https://doi.org/10.1063/1.5024783>

I. INTRODUCTION

Blood is a multiphase suspension with red blood cells and plasma being the two primary components. Microvessels carrying blood in the body continually bifurcate into smaller daughter vessels or merge into larger vessels. The action of cells being distributed into the daughter vessels as they pass through a bifurcation is typically referred to as cell partitioning. A major consequence of cell partitioning in physiology is the heterogeneous distribution of cells observed across microvascular networks (Popel and Johnson, 2005). Such heterogeneity is critical to the healthy functioning of the body as it impacts the transport of oxygen and nutrients to tissues.

Over the past decades, a rich body of work has grown in the literature studying cell partitioning at bifurcations using *in vivo*, *in vitro*, and theoretical approaches. Such studies have predominantly shown that RBCs usually do not distribute to the daughter vessels with the same proportion as the

blood flow. Owing to the discrepancy in cell and volume flow distributions, the term “disproportionate partitioning” is often used. This phenomenon has also been referred to as the Zweifach-Fung effect (Svanes and Zweifach, 1968 and Fung, 1973). *In vivo* studies, e.g., by Krogh (1921), Pries *et al.* (1989), and Ellis *et al.* (1994), among others, typically consider RBCs flowing within the microvascular networks of live animals, such as that in the cremaster muscle of a hamster (Klitzman and Johnson, 1982) or ear of a rabbit (Schmid-Schonbein *et al.*, 1980). Pries *et al.* (1989) performed a detailed analysis of RBC partitioning at arteriolar bifurcations and developed a quantitative description in terms of an empirical relationship between the RBC flux ratio and the blood flow ratio [see also Pries and Secomb (2005)]. Empirical relations have also been developed by Klitzman and Johnson (1982) and Gould and Linninger (2015) based on *in vivo* data, among others. *In vitro* studies usually involve RBCs or representative particles such as polystyrene microspheres, flexible rubber disks, or aluminum disks flowing within microfluidic bifurcations (Chien *et al.*, 1985; Roberts and Olbricht, 2006; Yang *et al.*, 2006; Sherwood *et al.*, 2014; Clavica *et al.*, 2016;

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and Shen *et al.*, 2016). Fenton *et al.* (1985) investigated the effects of hematocrit, diameter, local geometry, and cell properties for RBCs flowing in a bifurcating channel. They also developed an empirical relation between the RBC flux ratio and blood flow ratio, based on their *in vitro* study. Theoretical studies can be broadly classified into two categories: computational models that seek to predict the transport behavior of individual cells through bifurcations by modeling the cells as discrete objects [e.g., Audet and Olbricht (1987) and Barber *et al.* (2008)] and analytical models which consider a hematocrit concentration advected with the fluid streamlines (Enden and Popel, 1994). Computational models usually involve bifurcating tubes (Hyakutake and Nagai, 2015 and Wang *et al.*, 2016) or channels (Barber *et al.*, 2011; Li *et al.*, 2012; Shen *et al.*, 2016; and Bacher *et al.*, unpublished) and accurately resolve the cellular-scale details.

Various studies have shown that the disproportionate partitioning, in general, arises in two ways: a daughter branch receiving a higher flow fraction either receives an even higher RBC fraction or a lower RBC fraction [e.g., Schmid-Schonbein *et al.* (1980), Pries *et al.* (1989), Sherwood *et al.* (2014), Shen *et al.* (2016), and Clavica *et al.* (2016)]. The former is termed as *classical* partitioning and the latter as *reverse* partitioning. Shen *et al.* (2016) observed the reverse partitioning in a single bifurcation to occur as the result of a two-file-type cell pattern in the feeding vessel with a depleted core and outer layer. This led to the low flow branch receiving less plasma than the high flow branch, which explained the observed reverse partitioning. Schmid-Schonbein *et al.* (1980) and Pries *et al.* (1989) observed such reverse behavior as the result of disturbances caused by nearby upstream bifurcations, as was also observed in Sherwood *et al.* (2014). In addition to identifying such novel phenomena and quantifying the cell partitioning in terms of controlling parameters, studies have elucidated the underlying hydrodynamic mechanisms. Barber *et al.* (2008) and Doyeux *et al.* (2011) considered the mechanisms associated with the passage of one cell through a bifurcation. Opposing tendencies were identified that caused individual cell trajectories to deviate from the fluid streamlines. The most notable of which was an attraction tendency toward the low-flow branch of the bifurcation. Barber *et al.* (2011) considered the passage of two cells through a bifurcation. They specifically identified two general types of interaction dynamics, resulting in a tendency to create either a balance or imbalance in the hematocrit distribution to the daughter vessels. These and other computational studies generally tend to suggest that the individual cell trajectory through a bifurcation is affected by cell deformability, cell-to-vessel size ratio, initial cell location in the feeding vessel, as well as the geometric features of the bifurcation (Wang *et al.*, 2016; Hyakutake and Nagai, 2015; and Yin *et al.*, 2013).

Disproportionate cell partitioning has also been referred to in the literature as phase separation, and the two classic components which comprise such behavior have been termed *plasma skimming* and *cell screening* [e.g., Yan *et al.* (1991a) and Enden and Popel (1994)]. Plasma skimming is based on the advection of a concentration (i.e., hematocrit) profile through a bifurcation by the fluid velocity field and how the profile is partitioned to the daughter vessels. Thus,

an inherent assumption underlying this component is that the cells are volume-less point particles, which allows an analytical treatment of the phenomenon as considered, e.g., by Yan *et al.* (1991b) and Perkio and Keskinen (1983). Additionally, experimental studies have utilized a plasma skimming analysis to aid in the interpretation of the results [e.g., Fenton *et al.* (1985) and Pries *et al.* (1989)]. Cell screening, on the other hand, refers to the effects of the finite-size of the cells with respect to the vessel diameter. Because of the finite-size effects, cell deformation and cell-cell interactions come into consideration, which have been investigated in the aforementioned studies.

The aforementioned computational models on cell partitioning in single bifurcations have offered highly valuable insights. The next step is to consider cell partitioning in successive bifurcations as observed *in vivo* Cokelet (2011). It has been shown that upstream bifurcations necessarily have an impact on the partitioning behavior. Schmid-Schonbein *et al.* (1980) and Pries *et al.* (1989) each demonstrated this *in vivo*, where in the latter the proximity of an upstream bifurcation was shown to result in a notable asymmetry in partitioning. This was also demonstrated *in vitro* by such studies as Carr and Wickham (1990), Roberts and Olbricht (2006), and Sherwood *et al.* (2014). It appears, however, that computational studies investigating cell partitioning with RBC suspensions flowing through multiple sequential bifurcations have not been performed, to our knowledge.

We have recently developed a direct numerical simulation (DNS) technique for the cellular-scale modeling of the blood flow in geometrically complex and *in vivo*-like microvascular networks. Using this model, we present the first large-scale simulations of RBC partitioning in microvascular networks comprised of many sequential bifurcations. We provide results in terms of both time-averaged and time-dependent behaviors of cell partitioning (Secs. III A and III B). We identify specific cellular-scale mechanisms that cause such time-dependent behaviors and show how they cause the partitioning to oscillate between the diverse types over time (Sec. III C). We further identify the major factors giving rise to these cellular mechanisms. By considering the cell behavior at bifurcations, we provide a direct, simulation-based analysis of how the sequential nature of bifurcations results in a positively or negatively skewed hematocrit profile and how this skewness leads to specific partitioning types (Sec. III E). Then, we quantify the classic plasma skimming and cell screening components causing the observed disproportionate partitioning (Sec. III F). Characteristic mechanisms associated with each of these separate components are analyzed, and their significance is elucidated.

II. OVERVIEW OF SIMULATIONS

The numerical methodology has been described in Balogh and Bagchi (2017a). An extensive validation study is also presented therein for different components of the DNS technique. The present study is based on the data generated from the simulations described in our recent work (Balogh and Bagchi, 2017b). The full details of the simulations are presented therein, including comparisons with *in vivo* data and

discussions on the assumptions and limitations of the model. A brief overview is provided below.

The numerical methodology is based on the immersed boundary method (IBM), implemented in the context of a finite-volume/spectral fluid flow solver. This provides an efficient tool for the simultaneous consideration of geometrical complexity of the microvascular networks and the extreme deformation of the red blood cells, such as that which occurs *in vivo* during their passage through small capillaries. Three different microvascular networks are digitally constructed following published *in vivo* images (Zakrzewicz *et al.*, 2002; Cassot *et al.*, 2006; and Benedict *et al.*, 2011) and data (Lipowsky *et al.*, 1978 and Fung, 1996). Figure 1 shows the three networks considered. As is evident, the networks are geometrically complex, and they include bifurcating, merging, and winding vessels resembling an *in vivo*-like architecture. Starting with a feeding arteriole, the vessels in our networks sequentially bifurcate until the smallest diameter capillaries are reached. The vessels then successively merge to create larger vessels, eventually terminating in one outlet venule. The three networks together provide 138 vessels and 45 bifurcations in total. The vessel diameters and lengths range from 6 to 24 μm and 25 to 165 μm , respectively, while the overall volume simulated is about $1.6 \times 10^6 \mu\text{m}^3$ for each network. The number of pre-capillary bifurcations ranges up to 12, allowing us to

study the influence of the sequential nature of vascular bifurcations.

In total, twelve simulations are performed. For 9 simulations, the flow is driven by a pressure difference between the inlet and outlet specified as 0.3–1.0 Pa/ μm . The exact values and resulting flow rates are presented in Balogh and Bagchi (2017b). The pressures at the entrance to the inlet vessel and at the exit of the outlet vessels are chosen so that the average pressure loss per length over each network matches the minimum, mean, and maximum values observed *in vivo* (Fung, 1996). For the remaining 3 simulations, the mean flow rate in the main feeding artery is specified as $(2\text{--}4.6) \times 10^{-13} \text{ m}^3/\text{s}$. The exact values of the specified flow rates, the resulting inlet-outlet pressure drops, and their relation to the pressure boundary conditions were discussed in Balogh and Bagchi (2017b). The purpose of performing simulations using different boundary conditions was to capture a range of physiological conditions. The chosen boundary conditions result in average velocities and other pertinent hydrodynamic quantities in agreement with *in vivo* data. Additionally, time dependent fluctuations in flow quantities such as pressure loss and hydrodynamic resistance were also observed and discussed in Balogh and Bagchi (2017b).

The hematocrit in the main feeding artery is initially maintained at around 30%, and the subsequent RBC distributions naturally develop and evolve throughout the networks. In the absence of flow, the shape of each RBC is prescribed as the experimentally observed biconcave discocyte with an end-to-end distance of 7.8 μm , surface area 134.1 μm^2 , and volume 94.1 μm^3 (Fung, 1993). Before simulation data are recorded for analysis, an initializing simulation is performed. RBCs are initially placed randomly throughout the network, and then they deform as they flow through the network. When they reach the outlet, they are re-directed to the network inlet via a shifted periodicity. During the initialization simulation, cells are repeatedly added to maintain the hematocrit in the main feeding artery at around 30%. Once the network is filled with cells and a quasi-steady state is reached, simulation data are stored for analysis. The overall physical time during which data were stored is approximately 0.7 s.

Each cell is modeled as a viscous hemoglobin drop enclosed by a zero thickness (i.e., 2D) hyperelastic membrane. This model allows the cells to undergo large deformation by external forces. Under healthy conditions, the viscosity of plasma is close to that of water, and the hemoglobin viscosity is approximately 5 times greater. This ratio is the relevant parameter input into the simulations, with the viscosities of each of these phases taken to be 0.0012 and 0.006 Pa s, respectively (Fung, 1993). We model the membrane as possessing resistance to shearing, area dilation, and bending. The shearing deformation and area dilation are described following the strain energy function of (Skalak *et al.*, 1973). The bending resistance is modeled following Helfrich's formulation for bending energy (Zong-can and Helfrich, 1989). Specifically, an expression of the bending force density is used. The implementation details of the model can be found in Yazdani and Bagchi (2012). We note that Guckenberger and Gekle (2017) and Guckenberger *et al.* (2016) demonstrated large differences in accuracy between different numerical approaches to

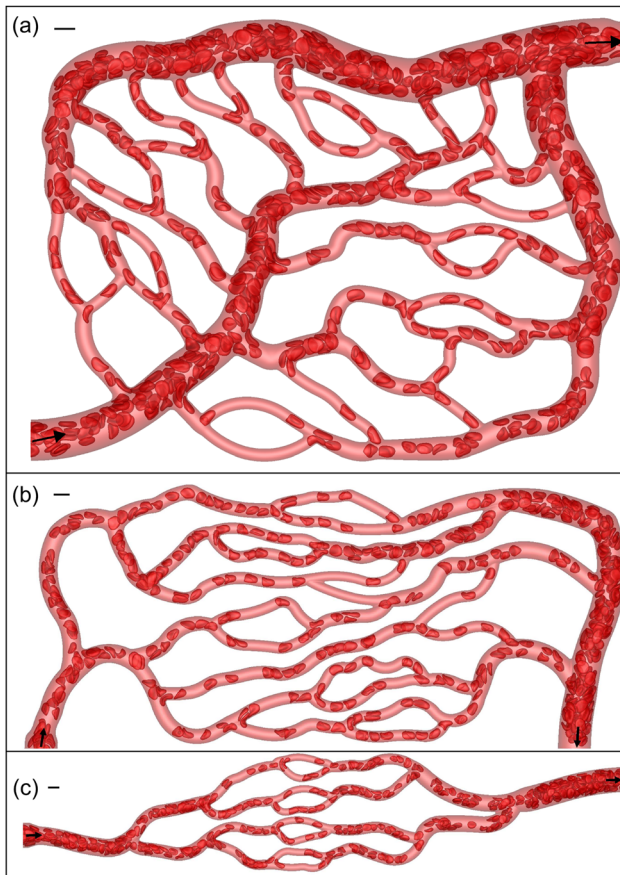


FIG. 1. Instantaneous visualizations from our simulations of the flow of red blood cells in each of the three microvascular networks (a)–(c). Inlet and outlet vessels are marked by arrows showing the flow direction. The scale bar (–) represents 10 μm .

compute bending forces. We have validated our model against experimental results of vesicle dynamics in the shear flow. Thus, we think our model is reasonably sufficient for the present study. The shear elastic and bending moduli are taken to be 5×10^{-6} N/m and 7×10^{-19} J, respectively. The higher bending modulus is used for numerical stability as it helps prevent mesh breakdown on the surface of the cells undergoing extreme deformation while squeezing through capillary bifurcations.

The governing equations for the fluid flow are the unsteady Stokes equations for a variable viscosity, constant density fluid, along with the conservation of mass,

$$\rho \frac{\partial \mathbf{u}}{\partial t} = -\nabla p + \nabla \cdot \mu [\nabla \mathbf{u} + \nabla \mathbf{u}^T], \nabla \cdot \mathbf{u} = 0. \quad (2.1)$$

The computational domain is discretized using a fixed, rectangular (Eulerian) mesh of about 8×10^7 points and resembles a box. Equation (2.1) is solved on this mesh using a projection method and a staggered arrangement of variables. In this approach, an advection-diffusion equation is first solved, followed by a Poisson-type equation to enforce the incompressibility condition. In the first step, all spatial derivatives are evaluated using second-order differencing, the body force terms are treated explicitly using a second-order Adams-Bashforth scheme, the viscous terms are treated semi-implicitly using the Crank-Nicholson scheme, and an alternating direction implicit (ADI) scheme is used. The Poisson equation is solved implicitly taking advantage of the periodicity of the computational domain which allows us to use the Fourier expansion for a fast solution. Further details can be found in Balogh and Bagchi (2017a). All interfaces, namely, the vascular walls and the cell membranes, are immersed within the Eulerian domain. The vascular walls are discretized using a triangular mesh that is comprised of about 3×10^6 elements. A sharp-interface ghost node IBM is used to implement the no-slip boundary condition on the vascular walls. In this method, a desired boundary condition is achieved by enforcing appropriate constraints on the fluid velocity at the Eulerian points that are immediately outside the vascular walls. The surface of each RBC is discretized by a Lagrangian mesh consisting of 5120 Delaunay triangles. A finite-element method is used to compute the membrane tensions resulting from the shearing deformation and area dilation. An expression of the bending force density derived from the bending energy is used (Zong-can and Helfrich, 1989). The coupling between the membrane and surrounding fluid is done by a continuous-forcing IBM. In this approach, the membrane forces are added to the Stokes equations as body forces that are spread over a finite region of the Eulerian domain using the 3D delta function. Any discontinuity across the membrane, such as the jump in viscosity, is also spread in a similar manner. The membrane velocity at each Lagrangian node on the cell surface is obtained from the surrounding Eulerian space using the same delta function. Knowing the membrane velocity, the surface nodes are advected to obtain the new locations and shapes of the cells. A lubrication-type repulsive force that varies as inverse distance cubed, in general accordance with lubrication theory, is used to prevent overlapping of cells with the vessel walls. With regard to overlapping between cells, this was not observed to occur in our simulations for the range of

hematocrit considered, likely due to the spreading of the membrane force over a finite Eulerian region that acts like a lubrication layer. The volume change for each RBC over the course of an entire simulation was calculated to be $<0.5\%$ on average; thus, no volume-conserving force was used.

Some limitations of our network model against actual physiological conditions may be noted. For example, the vessels in our networks have variable cross sections along their lengths although they are assumed non-deformable and circular in cross section, which is usually not the case *in vivo*. Additionally, the presence of the endothelial surface layer is neglected, which has been shown to influence the blood flow *in vivo* (Pries and Secomb, 2005). Furthermore, the layout of each of the simulated networks is planar in nature. While there is a physiological basis for this, for example, networks found in muscles (Bailey *et al.*, 2009) or in the retina (Li *et al.*, 2008), many *in vivo* networks have 3D topology. Data analysis and interpretation are, however, easier for the planar networks. Finally, this work does not consider leukocytes, which would likely have a considerable impact on the partitioning behavior. Since RBCs significantly out-number leukocytes in healthy blood (Fung, 1993), this work is thus representative of a period of time when leukocytes are not present in the part of the microvasculature of interest. Despite these limitations, the present DNS provides to-date the closest *in silico* model to actual *in vivo* geometry.

III. RESULTS AND DISCUSSION

A. RBC partitioning: Time-averaged results

First, the time-averaged behavior of cell partitioning is considered. Figure 2(a) presents the DNS data from all 45 bifurcations in the 12 simulations. Plotted here is the blood volumetric flow rate ratio (Q^*) between a daughter vessel and the feeding vessel of each bifurcation versus the erythrocyte flux ratio (N^*). These quantities are defined as

$$Q^* = \frac{\int_d \bar{\mathbf{U}} \cdot d\mathbf{A}}{\int_f \bar{\mathbf{U}} \cdot d\mathbf{A}}, \quad N^* = \frac{\overline{N_d(\tau)}}{\overline{N_f(\tau)}},$$

where the subscripts d and f refer to a daughter vessel and feeding vessel, respectively, $\mathbf{U}$ is the fluid velocity in the respective vessel cross section, N refers to the number of RBCs that passed through the bifurcation via the denoted vessel over a time window $\Delta\tau$ centered on time τ , and the over-bar denotes the time-averaging. The choice of $\Delta\tau$ is further discussed in Sec. III B.

As noted before, a hallmark of the microcirculatory blood flow is the disproportionate partitioning of red blood cells at vascular bifurcations. If the RBCs were to distribute in proportion to the blood flow, a linear relationship given by the straight line $N^* = Q^*$ would result. It can be observed in Fig. 2(a) that at most bifurcations in the simulations the disproportionate partitioning of RBCs is predicted, i.e., $N^* \neq Q^*$ for $Q^* \neq 0.5$. This is in general agreement with previous experimental findings.

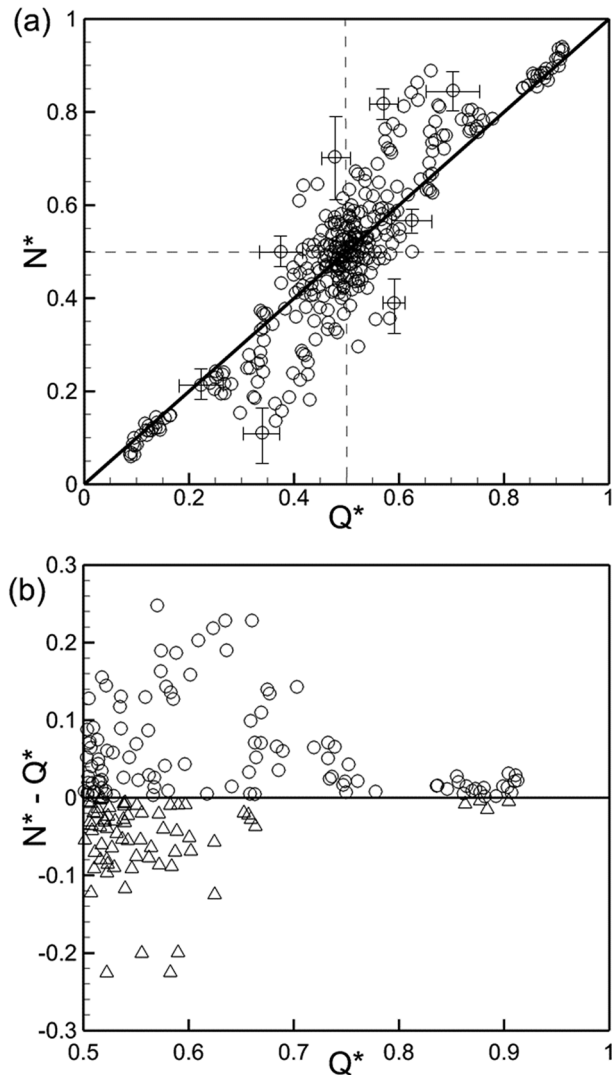


FIG. 2. (a) Time-averaged behavior of red blood cell partitioning at all bifurcations in the three networks simulated. The straight line represents $N^* = Q^*$, and the error bars give the RMS of the time history for a few representative data points. (b) The data are plotted as $N^* - Q^*$ versus Q^* to distinguish the different partitioning behavior. The data points (O) representing the *classical* partitioning are above the $N^* - Q^* = 0$ line. The data points (Δ) represent the *reverse* partitioning and are below the $N^* - Q^* = 0$ line. For clarity, we limit the abscissa range to 0.5–1.

As noted before, one way that the disproportionality manifests is for the daughter vessel that receives the greater fraction of the blood flow to receive an even greater fraction of the erythrocytes, i.e., $N^* > Q^*$ for $Q^* > 0.5$. Similarly then, due to the conservation of respective flow rates, the daughter branch that receives the lower fraction of the blood flow receives an even lower fraction of the RBC flux, i.e., $N^* < Q^*$ for $Q^* < 0.5$. This type of behavior is termed here as *classical* partitioning. It is evident from Fig. 2(a) that most of the bifurcations in the simulations exhibit this type. However, an opposite trend is also observed at a number of bifurcations which manifest as either $N^* < Q^*$ if $Q^* > 0.5$ or $N^* > Q^*$ if $Q^* < 0.5$. At such bifurcations, the daughter branch that gets the higher fraction of the flow actually gets a lower fraction of the RBCs or vice versa. This specific behavior, termed as *reverse* partitioning, has also been observed in prior studies [e.g., Schmid-Schonbein *et al.* (1980), Pries *et al.* (1989), Sherwood *et al.* (2014), and

Shen *et al.* (2016)]. To more clearly illustrate the two types of partitioning, we plot $N^* - Q^*$ versus Q^* in Fig. 2(b). As evident in this figure, for the bifurcations exhibiting the *classical* partitioning, $N^* - Q^* > 0$ and for the *reverse* behavior $N^* - Q^* < 0$.

It may be noted that Fig. 2 presents the aggregate of all data irrespective of different boundary conditions and network architectures used. Our objective in this work is not to study the influence of different boundary conditions or network architectures. Indeed individual RBC deformation and trajectories would be different for different boundary conditions and networks. Given the presence of many bifurcations in any simulated network, and the presence of many RBCs, the state of an individual cell at any time can be viewed as stochastic in the sense that it cannot be directly correlated to the boundary conditions or network architectures. Thus, the values of Q^* and N^* at a specific bifurcation in such networks vary under different boundary conditions without any specific trend. Therefore, it is appropriate to consider the aggregate data as presented in Fig. 2.

Furthermore, many of the aforementioned *in vivo* studies reported results for individual bifurcations over a large range of Q^* ($0 < Q^* < 1$). In such studies, Q^* was typically varied at each bifurcation by modifying the geometry by, for example, occlusion of a daughter vessel. Such data generally led to sigmoidal-type relationships between N^* and Q^* . Our goal here is not to develop such relations. Rather, we observe and study cell partitioning that naturally develops at each bifurcation in the simulated networks. That our simulation approach can predict the experimentally observed sigmoidal relationships has been demonstrated in Balogh and Bagchi (2017a).

B. Time-dependent RBC partitioning

To understand the means by which the time-averaged behaviors arise, we now examine the time-dependent results. We utilize the time-dependent flow ratio $Q^*(t)$ and RBC flux ratio $N^*(t)$ at each bifurcation to create plots analogous to Fig. 2(a). Such plots would represent how cell partitioning evolves in a time-dependent manner at a specific bifurcation. Before presenting this data, a few words are warranted. Considering the RBC flux in a time-averaged sense is analogous to the RBCs being treated as a continuum. In reality, RBCs are of finite size and flow as discrete objects in small vessels. The actual time-dependent signal for RBC flux appears as a binary-type signal that is not conducive to analysis. In order to get a useful time-dependent flux, the binary signal must be averaged over an appropriately chosen and small time window such that the resulting fluctuation frequencies match that of the whole blood flow rate. On average, these windows were on the order of 50–100 ms. This was selected based on the fact that the fluctuations in the whole blood flow rate are caused by the motion and deformation of the cells themselves. In testing this criteria, the time-signal for $N^*(t)$ derived from a specified window size was compared with visual observations of the simulations to ensure that what was being represented by this time signal was an accurate reflection of what was being observed with regard to time-dependent RBC motion. In general, it was observed that small changes to the window size (i.e., ± 20 ms) usually did not result in fundamentally different behaviors.

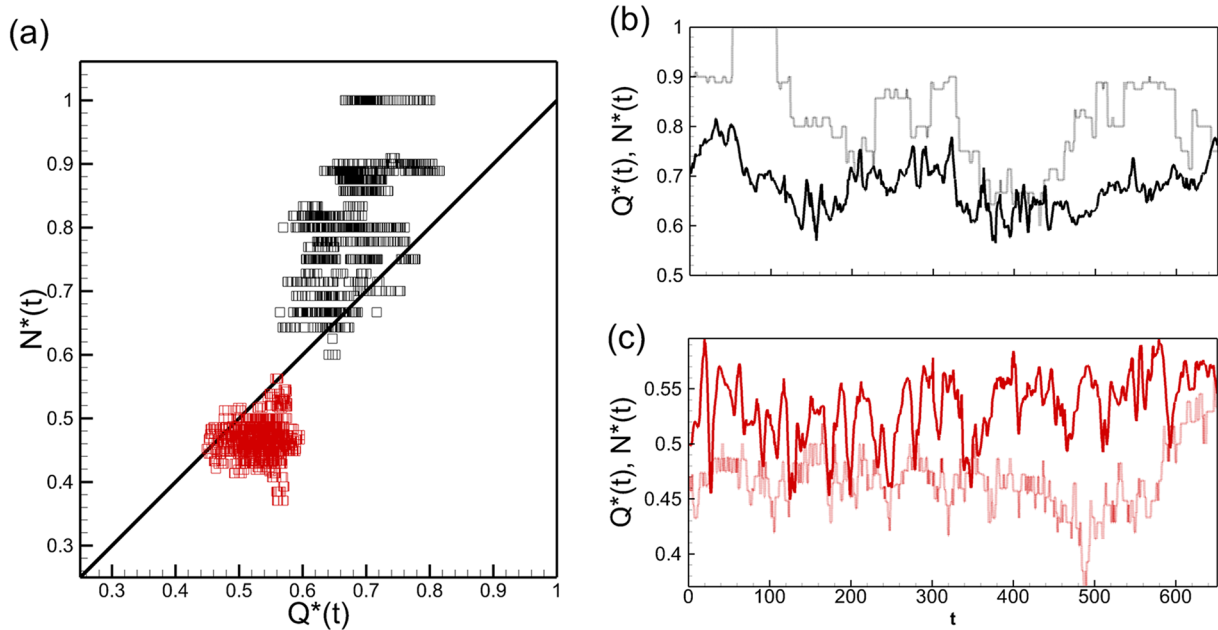


FIG. 3. Time-dependent cell partitioning. (a) $N^*(t)$ versus $Q^*(t)$ for two selected bifurcations. For one bifurcation, the average partitioning is the classical type (black symbols), while for the other it is the reverse type (red symbols). The straight line represents $N^* = Q^*$. [(b) and (c)] Time sequence of $N^*(t)$ (dotted line) and $Q^*(t)$ (solid line) for the two bifurcations (black and red), respectively. The classical partitioning occurs when $N^*(t) > Q^*(t)$ for $Q^*(t) > 0.5$ and vice versa, while the reverse type occurs when $N^*(t) < Q^*(t)$ for $Q^*(t) > 0.5$ and vice versa. Time in (b) and (c) is scaled by $\mu_0 a_0 / G_s$, where μ_0 is the plasma viscosity, a_0 is the undeformed RBC radius, and G_s is the membrane elastic modulus.

The time-dependent partitioning is shown in Fig. 3(a) by plotting $N^*(t)$ versus $Q^*(t)$ for two selected bifurcations and appears as a scatter plot. The time-averaged behavior in one bifurcation is of the classical type, while it is of the reverse type in the other. It is evident from this figure that while the time-averaged behavior could be either of the classical or reverse types, the time-dependent data are scattered above and below the straight line given by $N^*(t) = Q^*(t)$. This is also evident from the time sequences of $Q^*(t)$ and $N^*(t)$ shown in Figs. 3(b) and 3(c), where it can be seen that for each bifurcation considered there are periods in time when $N^*(t) > Q^*(t)$ for $Q^*(t) > 0.5$ and $N^*(t) < Q^*(t)$ for $Q^*(t) > 0.5$ and vice versa. Therefore, in a time-dependent sense, the cell partitioning at given bifurcation cycles between the classical and reverse types. The time-averaged behavior then depends on how frequently these time-dependent behaviors occur. Our results further suggest, and it has been verified from the simulation data, that the bifurcations that exhibit a proportional cell partitioning in a time-averaged sense (i.e., $N^* = Q^*$) also periodically oscillate between the classical and reverse type partitioning. The scatter plots additionally reveal that for either bifurcation under consideration, multiple values of $N^*(t)$ occur for the same $Q^*(t)$, and, conversely, the same value of $N^*(t)$ may occur for multiple values of $Q^*(t)$.

Most previous studies have considered cell partitioning in a time-averaged sense. Our result in Fig. 3 shows the dynamic nature of cell partitioning. It shows that a bifurcation which, on average, directs more cells to the high flow rate branch (i.e., classical partitioning) can have intermittent events of the opposite nature (i.e., reverse partitioning) and vice versa. Interestingly, Schmid-Schonbein *et al.* (1980) developed a 1D model for the capillary blood flow, and using a cell

distribution function determined from their *in vivo* data, they predicted time-varying disproportionate partitioning. However, it was not as a result of cell-cell and cell-vasculature interactions, but rather it was caused by the temporary increase in resistance associated with the passage of white blood cells (WBCs).

In general, it is observed that the time scales associated with fluctuation frequencies are related to both the cell residence time at bifurcations as well as the passage time of the cells through the vessels. The passage through the bifurcations, however, is observed to be dominant. This is because the events occurring at the bifurcations, such as the temporary RBC jamming, result in the most severe temporal fluctuations in the hydrodynamic quantities. On the other hand, the origin of the fluctuations coming from cell passage through the vessels is rooted in the temporary imbalances in resistance between the daughter branches. These primarily occur due to differences in cell distribution between the branches, as well as geometrical differences between the branches. Although less severe, these will also contribute. Thus, bifurcations that have similar occurrences of each of these contributing factors are also commonly observed to exhibit similar frequencies in fluctuations.

C. Cellular mechanisms

Next we investigate how instantaneous cell dynamics at bifurcations lead to a specific type of partitioning and also cause a distribution to oscillate in time between the classical and reverse types. In describing the underlying mechanisms, the terms “left” and “right” used below are meant to be with respect to the direction of the flow in a vessel (i.e., looking

downstream). Our main focus is on mechanisms which directly give rise to periods of reverse partitioning. By extension then, the behavior analogous to the inverse of the described mechanisms gives rise to periods of classical partitioning.

The first mechanism leading to the reverse type partitioning is presented in Fig. 4(a). For the bifurcation of interest, the RBCs flowing in the feeding vessel are skewed toward the right of the vessel. This skewed distribution arises because the way cells are partitioned at the upstream bifurcation. For the feeding vessel, the left side is plasma-rich, and the flow of cells lags that of plasma. The skewed distribution of the RBCs continues throughout the length of the feeding vessel. As a result, branch A receives a higher fraction of RBCs but a lower fraction of the blood flow, while branch B receives a lower fraction of RBCs but a higher fraction of blood flow, resulting in the reverse partitioning.

The second mechanism leading to reverse partitioning is shown in Fig. 4(b). The underlying cell dynamics for this mechanism results from a piling-up of the RBCs at the upstream bifurcation. In our simulations, we observe that RBCs can frequently pile-up at bifurcations, causing temporary and partial obstructions to the flow. Such piling (or lingering) of cells at bifurcations is discussed in Balogh and Bagchi (2017b). As a result of this behavior, RBCs enter the feeding vessel by moving around these temporary obstructions. For the time instant considered in Fig. 4(b), a temporary pile-up in the upstream bifurcation causes the cells entering the feeding vessel to be shifted to the left side. The skewed distribution continues throughout the length of the feeding vessel, and the flow of cells lags that of the plasma. As a result, branch A receives a higher fraction of cells but a lower fraction of the blood flow, resulting in the reverse partitioning. The snapshots in Fig. 4 are during time windows when reverse partitioning occurs. As noted before, the time-dependent partitioning can switch between the classical and reverse types, and the time-averaged partitioning depends on how frequently each occurs.

The mechanisms described earlier are essentially associated with an asymmetry of the RBC distribution over the cross section of the feeding vessel as a result of the specific ways the cells are partitioned at the bifurcation that is immediately upstream. Therefore, while these two mechanisms are related to specific hematocrit patterns in the feeding vessel,

they arise from the cellular interactions at an upstream bifurcation. Thus, in order to observe them, multiple bifurcations in sequence need to be considered as done in the present study. That upstream bifurcations can disturb the cell distributions entering a bifurcation downstream and cause reverse partitioning behavior has also been observed *in vivo*, e.g., by Pries *et al.* (1989) and Schmid-Schonbein *et al.* (1980) and *in vitro*, e.g., by Carr and Wickham (1990), Roberts and Olbricht (2006), and Sherwood *et al.* (2014).

Next we identify another mechanism that is directly related to cell-cell and cell-vasculature interactions. This type of interaction was previously identified in Barber *et al.* (2011) for a single, isolated bifurcation. They observed that either each RBC would enter a separate daughter vessel or each RBC would enter the same vessel, resulting in either a balance or imbalance in the hematocrit distribution. In our simulations, during periods when RBCs on average enter separate branches, the time-dependent partitioning behavior is observed to be closer to proportional. In contrast, when RBCs on average enter the same branch the disproportionality temporarily increases. This latter behavior is a component to the mechanism described next, and it typically arises at bifurcations associated with the smaller capillaries. This is illustrated in Fig. 5 for a selected bifurcation. First, we note that on average branch A gets 56% of the flow and 70% of the RBCs. Thus this bifurcation exhibits, on average, the classical partitioning. However, there are periods of time when branch A is congested with RBCs, but the flow rate is temporarily reduced due to the partial blockage of the branch by the cells, resulting in a reverse type partitioning. This is illustrated in Fig. 5(a). Also shown is the time-dependent RBC flux ratio $N^*(t)$ and blood flow ratio $Q^*(t)$ for branch A in Fig. 5(c), where this period of time is the first region shaded in gray. As evident in this figure, during this period, $Q^*(t) < 0.5$ but $N^*(t) > Q^*(t)$ in branch A. However, as branch A continues to be blocked, the flow resistance in this branch increases. The flow is then diverted to branch B. Subsequently the cells are drawn into branch B while branch A frees up, as shown in Fig. 5(b). As more cells are drawn to branch B, the resistance increases and the flow rate drops in that branch. At this moment, branch A receives a higher fraction of flow, but a lower fraction of cells, resulting in another time window of the reverse partitioning. This time window is marked in Fig. 5(c) by the second shaded region, and it is

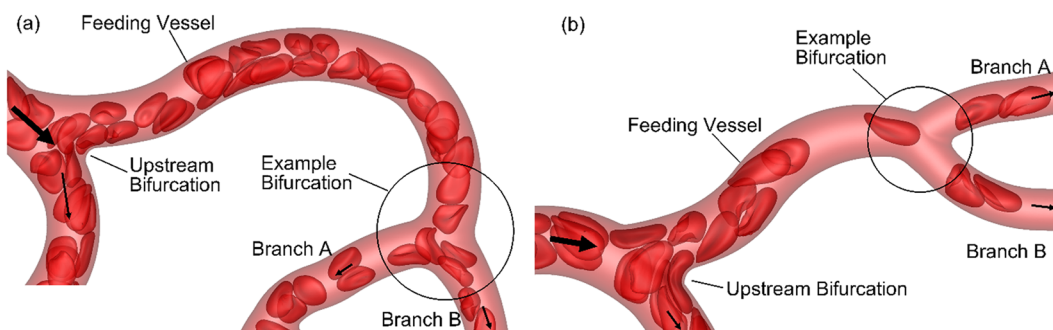


FIG. 4. Mechanisms leading to reverse partitioning at two bifurcations selected from the microvascular networks. The bifurcation of interest, the feeding and daughter vessels, and the upstream bifurcation are marked. The arrows indicate the flow direction. (a) The first mechanism in which cells in the feeding vessel are skewed to the right resulting in a higher RBC fraction entering into branch A. (b) The second mechanism in which a piling-up of cells at the upstream bifurcation causes other cells entering the feeding vessel to be shifted to the left side, and consequently, a higher fraction of RBCs for branch A.

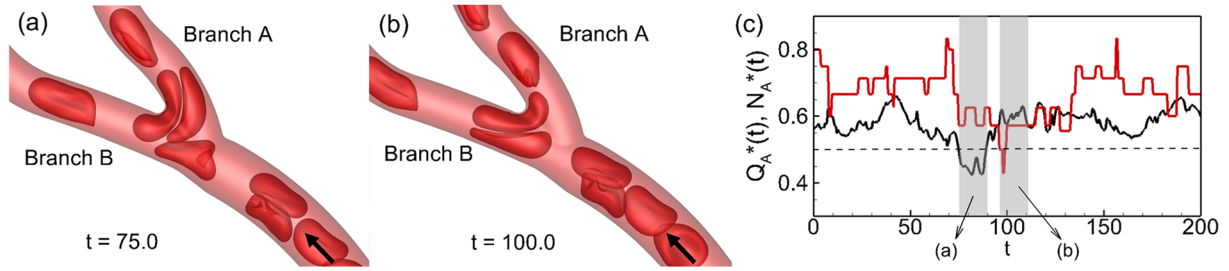


FIG. 5. Mechanism caused by cell-cell and cell-vasculature interactions in smaller capillaries leading to periods of classical and reverse type partitioning. A selected bifurcation from the simulated networks is considered. [(a) and (b)] Instances when cell partitioning is the reverse type. (c) Time history of blood flow ratio $Q^*(t)$ (black line) and RBC flux ratio $N^*(t)$ (red line) in branch A. The periods corresponding to (a) and (b) are marked in (c) using gray shades.

evident from the figure that for this period $Q^*(t) > 0.5$ but $N^*(t) < Q^*(t)$ in branch A. Outside these windows, the partitioning is the classical type, i.e., $N^*(t) > Q^*(t)$ when $Q^*(t) > 0.5$. In short, this specific mechanism is caused by temporary bursts of RBC flux that can occur in a branch as the direct result of cell-cell and cell-vasculature interactions. These in turn give rise to periods of classical and reverse partitioning behaviors.

These temporary bursts of RBC flux and their effect on the time-dependent partitioning are similar in spirit to that observed in Schmid-Schonbein *et al.* (1980). In their analytical model, such behavior was predicted due to a temporary increase in resistance due to the passage of a WBC through a single bifurcation. Here we show how this can occur with RBCs alone by considering the 3D cellular-scale details and interactions caused by complex network geometries. As previously mentioned, this action of RBCs entering the same daughter branch is also similar to that observed in Barber *et al.* (2011). Therein such behavior was termed either “herding” or “following,” with the distinction between the two dependent on differences in behavior between simulations with one cell or two cells. These interactions predominantly occurred only when the cells were initially close to each other in the feeding vessel. Since our simulations involve many cells passing through a bifurcation, the behavior similar to either “herding” or “following” can occur not only when the cells are close to each other in the feeding vessel but also when they are initially separated depending on the interactions involving other cells. At the same time, however, multiple cell-cell and

cell-vasculature interactions can also result in departures from the “herding” or “following” behavior.

The fourth mechanism we identify is a temporary flow-balancing mechanism. It provides a means for initiating a transition between different time-dependent partitioning behaviors at an individual bifurcation. It is directly associated with the aforementioned piling-up of RBCs that is observed to occur within bifurcations and temporarily causes the flow to be equally distributed between each daughter vessel. This predominantly occurs at the smallest capillary bifurcations where the confinement effects are most pronounced so that when the piling-up occurs the flow is simultaneously reduced into each branch and the flow distribution is temporarily equalized. As previously mentioned, one observation in Barber *et al.* (2011) involved each cell entering a separate daughter branch, which they termed a “trade-off” interaction. A byproduct of the fourth mechanism involves similar behavior in that following the temporary flow balancing we do observe that the RBCs can distribute in sequence to separate daughter branches. To some degree, this mechanism has also been observed in studies that used a continuum description of the blood flow, such as Kiani *et al.* (1994) or in Carr *et al.* (2005) where the result of the splitting of an idealized hematocrit slug at a bifurcation was shown to initiate a transition from one state to another.

Snapshots from two representative bifurcations at times when this mechanism is manifesting are shown in Figs. 6(a) and 6(b). The temporary congestion of cells within the bifurcation is evident, on account of the smaller vessel sizes causing

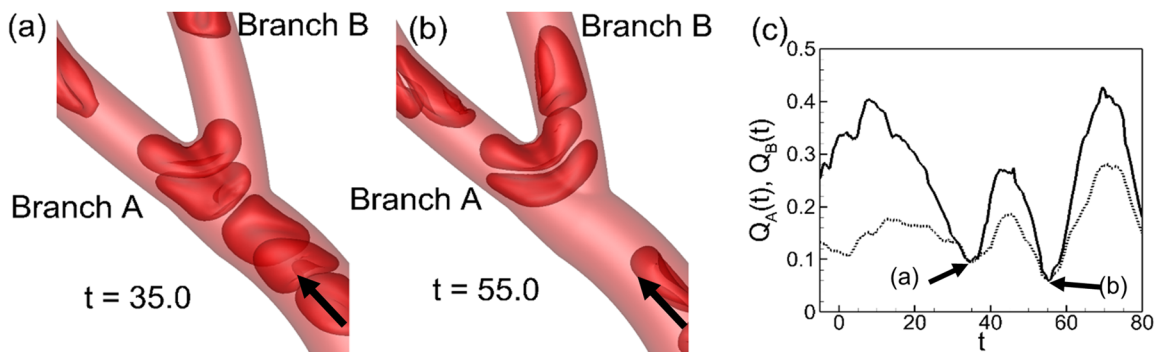


FIG. 6. Flow-balancing mechanism caused by RBCs piling up in smaller capillary bifurcations. A selected bifurcation from the simulated networks is considered. [(a) and (b)] Instances when temporary flow-balancing occurs. (c) Time history of the flow rates in branch A [$Q_A(t)$, solid line], and branch B [$Q_B(t)$, dotted line]. The periods corresponding to (a) and (b) are marked in (c).

TABLE I. Fraction of total time partitioning components is active at a few representative bifurcations. Data are provided for all bifurcations in the [supplementary material](#). The bifurcation type is listed as Capillary for terminal bifurcations on the arterial side and as Arterial for all others.

Bifurcation type	Classical partitioning	Reverse partitioning			
		Mechanism 1	Mechanism 2	Mechanism 3	Mechanism 4
Arterial	0.10	0.61	0.02	0.00	0.26
Arterial	0.70	0.00	0.30	0.00	0.00
Arterial	0.67	0.33	0.00	0.00	0.00
Arterial	0.38	0.02	0.09	0.16	0.35
Capillary	0.28	0.01	0.24	0.13	0.34
Capillary	0.43	0.03	0.04	0.24	0.26
Capillary	0.51	0.01	0.20	0.08	0.20
Capillary	0.11	0.02	0.00	0.17	0.61

a more pronounced confinement as the cells flow into the bifurcation. The time-dependent flow rates in each daughter vessel are shown in Fig. 6(c), and the points in time associated with these events are labeled. The temporary balancing of the flow is evident from the flow rate signals temporarily overlapping.

Note that the selected bifurcations in Figs. 4–6 provide representative examples. Each of the identified mechanisms is observed to occur, with varying frequency, at many bifurcations throughout the networks. While at a particular bifurcation one mechanism may predominantly occur more than another, the fraction of the total time is generally divided between each. This is illustrated for a few representative bifurcations in Table I, which gives a breakdown, for each bifurcation, of the fraction of total time each component is active. Outside of these mechanisms, the partitioning behavior is either of the classical disproportionate type or else it falls into a category not captured by the present classification scheme. Data are provided for all bifurcations in the [supplementary material](#), along with an additional discussion.

An interesting question arises as to whether the mechanisms described earlier can occur in single bifurcation models. As noted before, mechanisms 1 and 2 depend on what happens at an upstream bifurcation. So they would not be observed in a single bifurcation model unless the effect of the upstream bifurcation on the feeding vessel is somehow modeled. In contrast, mechanisms 3 and 4 were in fact observed in a single bifurcation model, as noted earlier. To appropriately compare the probability of either mechanism occurring in a single bifurcation model, the issue of boundary conditions will come into play. With the network simulations, the velocity and pressure fields are not fixed at the bifurcation inlet/outlet, but rather they arise naturally and are highly time-dependent. This affects the means by which these mechanisms manifest and would have to be considered when selecting the boundary conditions to prescribe in a single bifurcation model. Any fixed pressure or flow boundary conditions may bias the results being compared and cause departures from the naturally developing conditions captured in the network simulations.

Concerning the nature of these mechanisms, a point of distinction is that these are not meant to be predictive in nature but are really more observational. To some extent, their

existence depends on subtle details in the complex cell-cell and cell-vasculature interactions which cause the mechanisms to arise. That they are limited to being observational in nature partly speaks to the fact that these details are not yet well understood. In reality, these mechanisms are really just a first step in understanding how certain partitioning behaviors can arise. The next step would then be to understand their nature to the degree that their occurrence can be predicted. What is understood though is that these subtle details are, to a large extent, stochastic in nature. This was discussed in our recent work ([Balogh and Bagchi, 2017b](#)). Related to the present work then, the temporary patterns form as the result of a very complex sequence of interactions. These eventually culminate in the collective RBC behavior which forms the basis for the identified mechanisms. At present then, while we can identify when these mechanisms have occurred, as mentioned, this is not yet something predictive.

D. Influence of vessel diameter

The quantity $|N^* - Q^*|$ can be thought as a measure of the degree of disproportionality in cell partitioning. Of interest is the dependence of $|N^* - Q^*|$ on the feeding vessel diameter

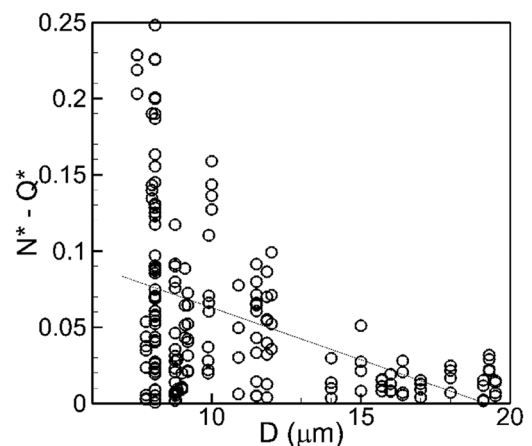


FIG. 7. Influence of the feeding vessel diameter on the degree of disproportionality in cell partitioning defined as $|N^* - Q^*|$. Symbols are DNS results, and the line is a fit through the data.

which is shown in Fig. 7. It is observed that on average $|N^*-Q^*|$ is diameter-dependent, and its value increases with decreasing diameter. What this means is that as the vessel diameter becomes smaller, cells on average are partitioned even more disproportionately. This result underscores the influence of the finite size of the RBCs. However, as is evident from Fig. 7, the range of values of $|N^*-Q^*|$ also increases with decreasing diameter. This means that very different types of partitioning can occur at vessels with smaller diameters—from a highly disproportionate partitioning corresponding to a large value of $|N^*-Q^*|$ to a nearly proportional partitioning corresponding to $|N^*-Q^*| \rightarrow 0$. It also tends to explain why some capillary vessels *in vivo* have high hematocrit while others may be nearly devoid of cells (Lipowsky *et al.*, 1978; Fung, 1996; and Pries *et al.*, 1996). One possible mechanism by which a large disproportionality can occur is the reverse type partitioning described previously in Fig. 5. With this mechanism, one daughter vessel at a bifurcation can become congested with RBCs while the flow is significantly reduced, resulting in a large value of $|N^*-Q^*|$. Other possible mechanisms are those described in Fig. 4 and could be associated with either the classical or reverse partitioning. Under these mechanisms, cells in the feeding vessel are predominantly shifted to one side, which leads to a large positive or negative value of N^*-Q^* . In contrast, a small value of $|N^*-Q^*|$ may occur when the cells in the feeding vessel are uniformly distributed and thus leads to a more proportional partitioning to the flow.

It may be noted that previous studies have shown the importance of the ratio of daughter vessel diameters. For example, in the empirical correlation developed by Pries *et al.* (1989), there is an asymmetry parameter that is dependent on the diameter ratio. The effect of this ratio becomes most pronounced when N^* can be plotted over a large range of Q^* for one bifurcation to develop an N^*-Q^* curve. This can be obtained, for example, by gradually occluding a daughter vessel, as was done in this previous work (see also Sec. III A). The present simulations, however, do not permit modeling such vessel occlusion. As such, it is difficult to draw any conclusions as to the effects of the diameter ratio on the distribution behavior based on our DNS results. Such a dependence on the diameter ratio was in fact examined; however, this did not offer anything conclusive with regard to distribution asymmetry or other features.

E. Influence of hematocrit skewness

In Sec. III C, we discussed the mechanisms associated with cell partitioning. The general picture that arises from the two underlying mechanisms presented in Fig. 4 is that an upstream bifurcation causes a skewed distribution of RBCs in a vessel, leading to the instantaneous reverse partitioning at a downstream bifurcation. The skewness in the cell distribution may alter from one side of the feeding vessel to another due to the cell dynamics at the upstream bifurcation, also causing the downstream cell partitioning to switch between the classical and reverse types. These instantaneous events ultimately comprise the time-averaged partitioning. Therefore, one way to investigate the influence of upstream bifurcations on cell partitioning is to consider the time-averaged hematocrit and velocity profiles in the feeding vessels.

For this analysis, the separation surface appears as a relevant quantity, which has been used in prior studies studying cell distributions [e.g., Schmid-Schonbein *et al.* (1980), Chien *et al.* (1985), and Pries *et al.* (1989)]. This surface represents the boundary in the feeding vessel separating regions of the flow that go into each daughter vessel. For each bifurcation, it is obtained *a posteriori* from the simulation data by considering a dense collection of streamlines determined from the time averaged velocity field, moving through the bifurcation. Starting at a cross section at the bifurcation inlet, the streamlines distribute into either of the two daughter vessels, thus forming two separate groups. The separation surface is given by the interface dividing these two groups. This surface for a selected bifurcation is shown in Fig. 8. It may be noted that the separation surface, in general, is not a flat surface but rather a curved one as evident from this figure.

First, reverse partitioning is presented in Fig. 8, where a part of a simulated network is shown that includes the selected bifurcation of interest, the feeding vessel, and the upstream bifurcation. The time-averaged hematocrit distribution throughout the network segment is shown in Fig. 8(a). Also shown are the hematocrit and velocity profiles at one location in the feeding vessel and the separation surface. The velocity and hematocrit distributions over the cross section of the feeding vessel at this specific location are also shown in Figs. 8(b) and 8(c) along with the separation surface. Since the way cells are partitioned at the upstream bifurcation, the hematocrit profile in the feeding vessel is skewed to the left

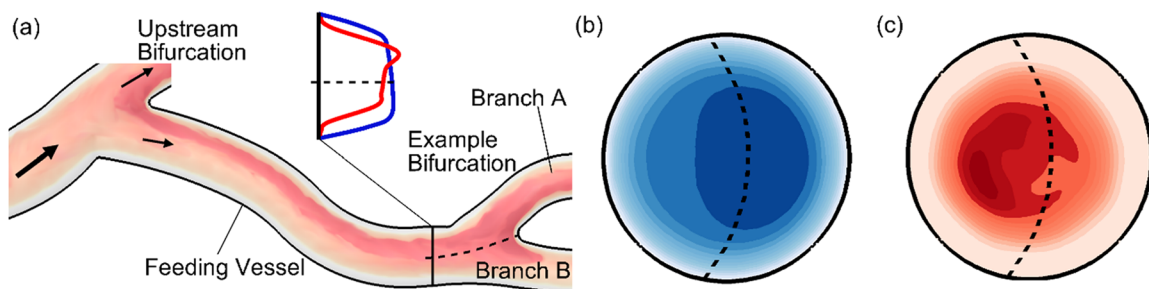


FIG. 8. The mismatch between hematocrit and velocity skewness in the feeding vessel leading to the reverse type partitioning. (a) A selected bifurcation is shown. Color contours represent hematocrit distributions (darker representing higher values). Also shown are the separation surface (dashed line), hematocrit (red line), and velocity (blue line) profiles at one location ahead of the bifurcation. The cross-sectional distributions of velocity and hematocrit at this location are shown in (b) and (c), respectively. The dashed line in (b) and (c) represents the separation surface.

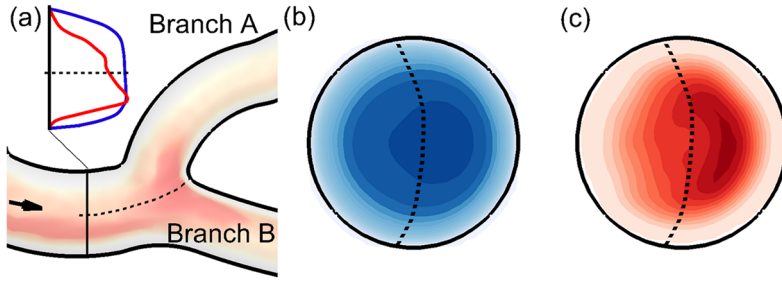


FIG. 9. The classical partitioning occurs when the hematocrit and velocity profiles in the feeding vessel are skewed to the same side of the separation surface. Contours and line symbols are the same as those in Fig. 8.

of the separation surface. The cells, however, lag behind the flow, resulting in the whole blood velocity profile skewed to the right of the separation surface. The mismatch between the velocity and hematocrit distributions with respect to the separation surface is also evident in Figs. 8(b) and 8(c). As a result of the mismatch, daughter branch A receives a higher fraction of the cells but a lower fraction of the flow, leading to the reverse type partitioning. A connection between the feeding vessel concentration profiles and reverse partitioning behavior was also observed in Shen *et al.* (2016). As previously mentioned, they explained this reverse behavior to occur as the result of a specific profile in the feeding vessel leading to the low flow branch recruiting less plasma than the high flow branch.

For the classical partitioning, a representative example is shown in Fig. 9. Here the velocity and hematocrit profiles are both skewed to the right side of the separation surface in the feeding vessel. Hence, daughter branch B gets the higher fraction of the blood flow as well as the higher fraction of the RBC flux, resulting in the classical partitioning.

To further quantify the cell partitioning in terms of the flow and hematocrit profile mismatch, we compute the skewness in the hematocrit profile at a location in the feeding vessel ahead of the bifurcation following Sherwood *et al.* (2014),

$$S_h = \text{sign} \left\{ \frac{\int_{S_{1/2}} U(r, \theta) dA}{Q_0} - 0.5 \right\} \times \left| \frac{\int_0^r \int_{\theta_1}^{\theta_1+\pi} H_t(r, \theta) dA}{\int_0^r \int_{\theta_1}^{\theta_1+2\pi} H_t(r, \theta) dA} - 0.5 \right|, \quad (3.1)$$

where U and H_t are the time-averaged blood velocity and hematocrit over the vessel cross section at the chosen location, $S_{1/2}$ is the cross-sectional area on one side of the separation surface, and θ_1 to $\theta_1 + \pi$ is the part of the circumference that lies on the same side. With the skewness defined in this manner, the value gives the fraction of the profile residing on one side of the vessel centerline, while the sign specifies which side. The skewness is positive if the hematocrit profile is skewed toward the daughter vessel that gets more flow and vice versa. Figure 10 shows N^*-Q^* versus S_h for all bifurcations. It is seen here that for most bifurcations, $N^*-Q^* > 0$ if $S_h > 0$ and vice versa. Thus, on average, a positive skewness in the hematocrit of the feeding vessel is associated with the classical

partitioning, while a negative skewness is associated with the reverse one.

This analysis further demonstrates how the sequential nature of bifurcations in a vascular network is a key component causing the diverse types of cell partitioning. It has been observed *in vivo* (Carr, 1989 and Carr and Xiao, 1995) that the length between successive bifurcations is typically not sufficient to discount the influence of upstream bifurcations. That such features may affect the partitioning behavior has been noted by a number of *in vivo* and *in vitro* studies (Schmid-Schonbein *et al.*, 1980; Pries *et al.*, 1989; Carr and Wickham, 1990; Roberts and Olbricht, 2006; and Sherwood *et al.*, 2014). Here we show, through our *in silico* models, how in physiological-type geometries there is a definitive and quantitative connection between cellular events at upstream bifurcations and the partitioning behavior downstream.

F. Plasma skimming and cell screening

To further elucidate the underlying hydrodynamics of disproportionate partitioning, we focus now on the two classical components, namely, plasma skimming and cell screening, as defined in such studies as Yan *et al.* (1991a) and Enden and Popel (1994). As previously mentioned, plasma skimming is based on the advection of a hematocrit concentration by the fluid velocity field and thus assumes that the cells

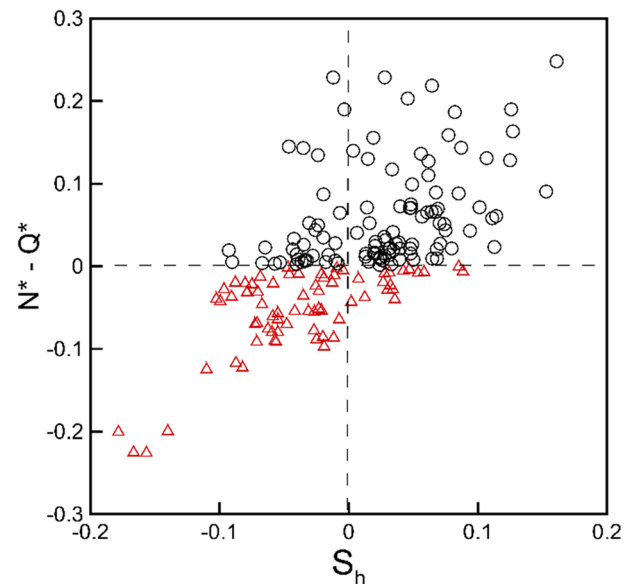


FIG. 10. N^*-Q^* versus skewness S_h . Black circles represent the classical partitioning, and red triangles represent the reverse type.

are volume-less point particles, while cell screening accounts for the finite-size of the cells. Furthermore, plasma skimming is a steady-state phenomenon, while cell screening is affected by the time-dependent cell behavior in bifurcations. In Secs. III F 1 and III F 2, we present quantitative results on each component as obtained from the simulations.

1. Plasma skimming

To isolate the plasma skimming component, we consider the advection of a hematocrit profile through the modeled networks. The hematocrit profile is specified at the inlet to a network and advected by the fluid as a scalar field as

$$\frac{\partial H_t}{\partial t} + \mathbf{U} \cdot \nabla H_t = 0, \quad (3.2)$$

where $\mathbf{U}$ is the time-averaged velocity field obtained from the full DNS results (i.e., simulations with the RBCs). The inlet hematocrit profile is taken to be the time-averaged hematocrit profile at that location obtained from the DNS. The advection equation is numerically solved until a steady hematocrit distribution is established. Then the ratio N^* for each bifurcation is computed as the ratio of the hematocrit flux $\int H_t \mathbf{U} \cdot d\mathbf{A}$ in a daughter branch to that in the feeding vessel.

The steady-state hematocrit distribution from the pure plasma skimming simulation for one network is provided in Fig. 11(a). The resulting N^* values, denoted as N^*_{PS} , are presented in Fig. 11(b) for a few representative bifurcations. As evident here, our simulations predict phase separation by the plasma skimming mechanism alone. The phase separation due to plasma skimming alone occurs due to a non-uniform hematocrit profile. If the hematocrit distribution at the inlet to a bifurcation was uniform over the entire cross section, it would be partitioned proportionately to the flow, and the phase separation would not occur. In the plasma skimming simulations, the hematocrit profile at the inlet to the vascular networks is non-uniform. As the profile is advected, the hematocrit distribution naturally develops in the network, and the resulting hematocrit profiles ahead of each bifurcation, in general, are also non-uniform.

Also shown in Fig. 11(b) are the results from the full DNS simulations. Two important observations are made here. First, the plasma skimming component alone cannot account for the full extent of disproportionate partitioning. Rather, it under-predicts the amount of disproportionality. That is, the

data points representing the pure plasma skimming component are closer to the $N^* = Q^*$ line, while the actual DNS data are further away. In almost 80% of the bifurcations, plasma skimming is observed to under-predict the disproportionate cell partitioning. To provide a more general picture of this trend, data for all bifurcations are plotted in Fig. S2 of the [supplementary material](#).

Second, both the classical partitioning and reverse partitioning are present in the results of the pure plasma skimming simulation. This result directly follows from Sec. III E, where it was observed that the skewness in the hematocrit profile on average dictates the partitioning behavior at each bifurcation. Since the H_t distribution in the plasma skimming simulations is allowed to naturally develop through each sequential bifurcation, the profiles at some bifurcations could be either positively or negatively skewed with respect to the fluid velocity, resulting in the classical or reverse partitioning.

The conclusions that the plasma skimming component under-predicts the total cell partitioning, and that it can yield the reverse type partitioning, are robust. We have done additional simulations of plasma skimming by isolating each individual bifurcation in the networks. In such simulations, a “local” hematocrit profile that is obtained from the time-averaged DNS data is specified at the inlet to the bifurcation. The profile is advected through the selected bifurcation, and the value of N^* is obtained. Plasma skimming based on such local profiles also resulted in an under-prediction of the total cell partitioning, as well as the reverse partitioning. Furthermore, if the advecting fluid velocity $\mathbf{U}$ is taken to be the velocity of pure plasma without any RBCs flowing through the network, the same conclusions about the plasma skimming mechanism are drawn.

The plasma skimming contribution to the overall disproportionality in cell partitioning can be quantified as $N^*_{PS} - Q^*$. The magnitude of $N^*_{PS} - Q^*$ is plotted in Fig. 12(a) as a function of feeding vessel diameter. The figure shows that the plasma skimming contribution increases with decreasing diameter. At first glance, this is a surprising result because plasma skimming is based on the assumption of point particles following the fluid streamlines. This diameter dependency arises from the bluntness in the hematocrit profile. To investigate this, we compute the bluntness index (B_H) of the hematocrit profile as obtained from the pure plasma skimming simulations. Mathematically, B_H is written as

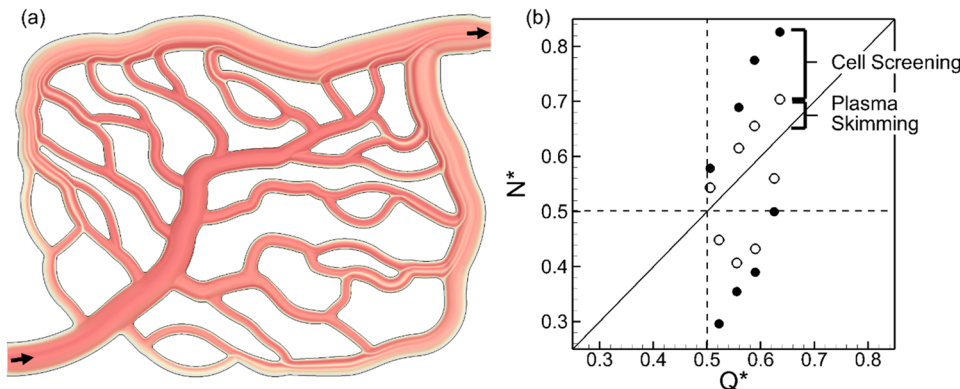


FIG. 11. (a) Simulation of pure plasma skimming in the network shown in Fig. 1(a). Color contours represent H_t distribution at steady-state. (b) N^* versus Q^* plot showing the results of the plasma skimming simulations (o). Also shown are the data from the full DNS simulations (i.e., with the RBCs) ($\bullet$). Only a few data points are shown to avoid clutter.

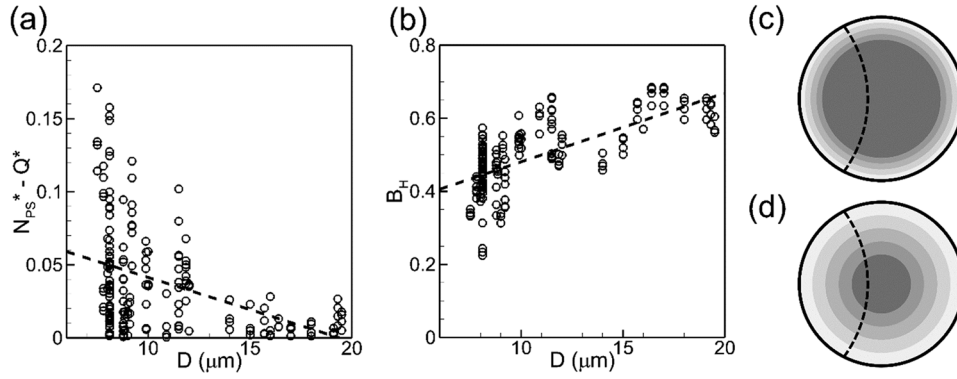


FIG. 12. (a) Contribution of plasma skimming to overall disproportionality in cell partitioning is considered by plotting $|N_{PS}^* - Q^*|$ versus feeding vessel diameter. Here N_{PS}^* is the tracer flux ratio obtained from the pure plasma skimming simulations as noted in Fig. 11. (b) Bluntness index for the hematocrit profiles in the feeding vessel. The straight lines in (a) and (b) are a fit through the data. [(c) and (d)] Illustrate the influence of hematocrit bluntness on the degree of disproportionality in cell partitioning. The dashed curves represent the separation surface.

$$B_H = 2 \cdot \left(1 - \frac{\int_0^{2\pi} \int_0^{\sqrt{0.5}R} H_t dr d\theta}{\int_0^{2\pi} \int_0^R H_t dr d\theta} \right) \quad (3.3)$$

following Sherwood *et al.* (2014). Thus, B_H provides a measure of how focused the hematocrit profile is toward the vessel centerline: $B_H = 1$ if the H_t profile is uniform over the cross section and $B_H = 0$ if the profile is focused. As can be seen in Fig. 12(b), B_H on average increases with decreasing D . This observation is in agreement with the plasma skimming calculations performed by Pries *et al.* (1989) based on their *in vivo* data. It means that, on average, the hematocrit profile is more focused in vessels with smaller diameters. A focused profile can lead to an increased disproportionality in cell partitioning. This is schematically illustrated in Figs. 12(c) and 12(d). For the H_t distribution that is more focused [Fig. 12(d)], there is a greater imbalance of hematocrit between each side of the separation surface compared with the distribution that is more blunt [Fig. 12(c)].

Figure 12(a) also suggests that there is a wider variability of the plasma skimming contribution at the vessels with smallest diameters. This trend also follows from the bluntness of the hematocrit profile and can be illustrated by Figs. 12(c) and 12(d). For the focused distribution in Fig. 12(d), a small change in the location and shape of the separation surface can result in a relatively large change in N_{PS}^* . In contrast, for a distribution that is more uniform as in Fig. 12(c), N_{PS}^* is less sensitive to such small changes. Thus, a focused hematocrit profile, which on average is more frequently observed in smaller diameter vessels, leads to a wider range of disproportionality.

2. Cell screening

The difference between the DNS result and the plasma skimming result is the overall cell screening effect. In Sec. III F 1, it was shown that at most bifurcations plasma skimming under-predicts the phase separation. This leaves cell screening to make up the difference. This relationship was apparent in Fig. 11(b). Quantitatively, the cell screening contribution can be computed as $N_{CS}^* = N_{DNS}^* - N_{PS}^*$ for a given Q^* , where the subscript CS refers to cell screening. The magnitude of N_{CS}^*

is presented in Fig. 13 as a function of feeding vessel diameter. This figure, together with Figs. 11(b) and 12, shows that both cell screening and plasma skimming are of similar magnitudes and that together they contribute to phase separation. Additionally, Fig. 13 shows that the magnitude of cell screening increases with decreasing vessel diameter on account of the finite cell size.

Next we investigate the mechanisms giving rise to cell screening. We observe two primary mechanisms to be the main contributors as discussed below.

The first mechanism underlying the cell screening component is primarily associated with the average cell behavior, and we term it as the *all-or-none* effect. Note that when considering plasma skimming, a hematocrit profile is considered that is the result of the aggregate effect of many RBCs passing through the vessel over a period of time. Thus the influence of each RBC is spread over a region, figuratively resulting in a fraction of each RBC going into each of the daughter vessels at a bifurcation. In contrast, due to the finite nature of each RBC, each daughter vessel gets either “all or none” of the RBC. This increases the degree of disproportionality in cell partitioning. This mechanism is illustrated in Fig. 14 where selected bifurcations from the simulated networks are

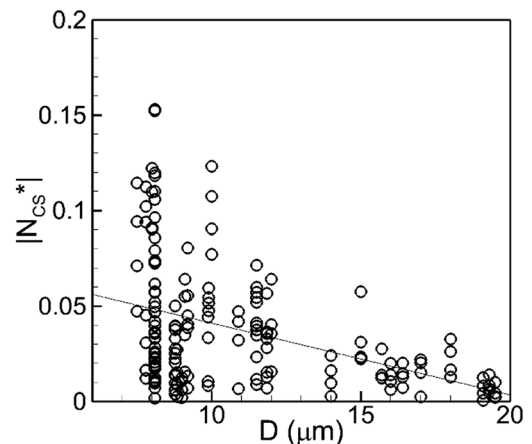


FIG. 13. Magnitudes of cell screening contribution computed as $|N_{DNS}^* - N_{PS}^*|$ versus feeding vessel diameter. The line is a fit through the data.

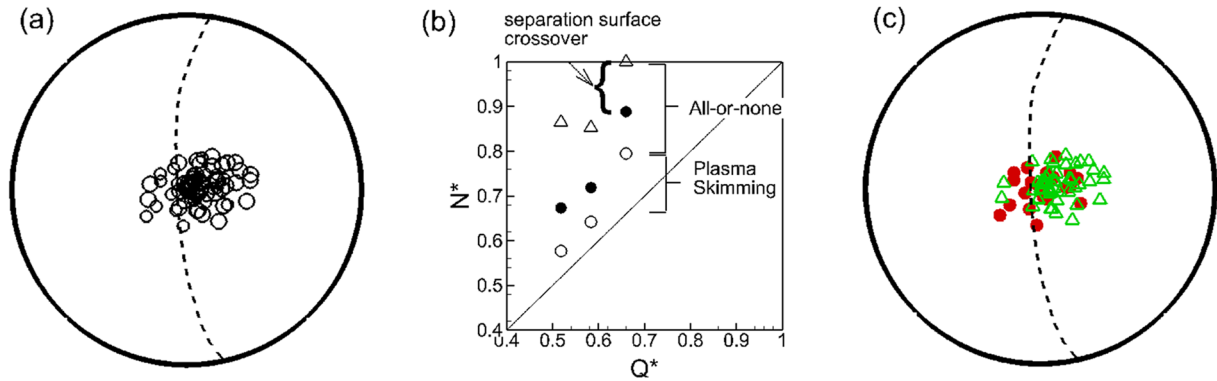


FIG. 14. Dominant mechanisms of the cell screening component. (a) The “all-or-none” effect. Centroids of the RBCs are shown over a cross section in the feeding vessel of a selected bifurcation. The dashed line represents the separation surface. (b) Some representative data points showing the “all-or-none” effect. $\circ$ represents results based on plasma skimming alone, $\bullet$ represents DNS results, and Δ represents results based on combined plasma skimming and the “all-or-none” effect. (c) “Separation surface crossover” mechanism. Data in (a) are re-plotted by identifying the RBCs that enter the left and right daughter vessels by red circles and green triangles, respectively. The contribution from this mechanism is also marked in (b).

considered. Figure 14(a) shows the centroid location of each RBC on a cross-sectional plane in the vessel feeding the bifurcation. Also shown is the separation surface line.

According to the “all-or-none” effect, every RBC whose centroid is on one side of the separation surface should enter into the daughter vessel on that side. Thus the RBCs whose centroids are on the left side of the separation surface should enter the left daughter branch and vice versa. In contrast, under pure plasma skimming, each daughter vessel would get a “fraction” of each RBC. Thus, the “all-or-none” portion of the cell screening component increases the degree of disproportionality in cell partitioning over and above that predicted by pure plasma skimming. The value of N^* resulting from the “all-or-none” effect can be obtained as the ratio of the number of RBC centroids on one side of the separation surface to that of the total number of RBCs. For example, for the bifurcation in Fig. 14, the plasma skimming component predicts $N^*_{PS} = 0.62$ for the daughter vessel lying on the right side of the separation surface. In contrast, the “all-or-none” mechanism results in $N^* = 0.87$, thereby significantly enhancing the disproportionality. Additional data shown in Fig. 14(b) also support the increased disproportionality in cell partitioning by the “all-or-none” mechanism.

The “all-or-none” effect can be thought as plasma skimming with finite-sized cells. In reality, however, cells do not follow the streamlines. To this end, the second mechanism underlying the cell screening component is the *separation surface crossover*. It is evident from Fig. 14(b) that the DNS data yield, on average, lower values of N^* than the combined effect of plasma skimming and the “all-or-none” mechanism. This can happen if the cells whose centroids are on one side of the separation surface enter the daughter vessel that is on the other side. We replot the data of Fig. 14(a) in Fig. 14(c) by identifying the cells that have crossed the separation surface. We find that for this bifurcation 30% of the RBCs in the feeding vessel have crossed the separation surface before entering the daughter vessels. It is evident that this component of cell screening reduces the extreme disproportionality predicted by the all-or-none component.

What causes the cells to cross the separation surface? We find two primary means by which this happens, each

associated with the time-dependent RBC dynamics. The first is the direct result of cell-cell and cell-vasculature interactions. This generally occurs when a cell straddles around a bifurcation partly blocking a daughter vessel and forcing another RBC to enter the other daughter vessel. Figure 15(a) provides a time sequence illustrating the process in a selected bifurcation. In this figure, an RBC can be observed to straddle around the bifurcation, partly blocking the left daughter branch. This forces the following RBC that was on the left of the separation surface to cross over and enter the right daughter branch.

The second means by which cells cross the separation surface is associated with the steady-state nature of the surface as opposed to the unsteady nature of cell dynamics. The separation surface is computed in a time-averaged (or steady-state) sense, while the cell crossing is an unsteady process. If the majority of the cells at a bifurcation flow into one daughter vessel for a certain period of time, the resistance in that daughter vessel would increase. The flow would then be diverted into the other daughter vessel taking with it some of the RBCs flowing in the feeding vessel. As such, these RBCs would enter the daughter vessel by crossing the separation surface.

How important is the “separation-surface-crossover” mechanism? To answer this, we compute the ratio of the number of RBCs that enter a daughter vessel by crossing the separation surface to the total number of RBCs in the feeding vessel of a bifurcation. This ratio, denoted by η_{ss} , is presented in Fig. 15(b) as a function of feeding vessel diameter. As is evident, the crossover fraction η_{ss} increases with decreasing vessel diameter on account of the finite-size effect. More importantly, the value of η_{ss} is quite large and as high as 0.45 in the smallest vessel. Thus, in some bifurcations associated with the smallest capillaries, nearly 50% of the RBCs in the feeding vessels are observed to cross the separation surface. In Fig. 15(c), we present η_{ss} as a function of the daughter vessel diameter. This figure gives the fraction of cells that have entered a particular daughter vessel by crossing the separation surface. It shows that there are bifurcations where 100% of the RBCs have entered the respective daughter vessel by crossing the separation surface.

The above result is physiologically important. If the separation surface crossover did not occur, some daughter vessels

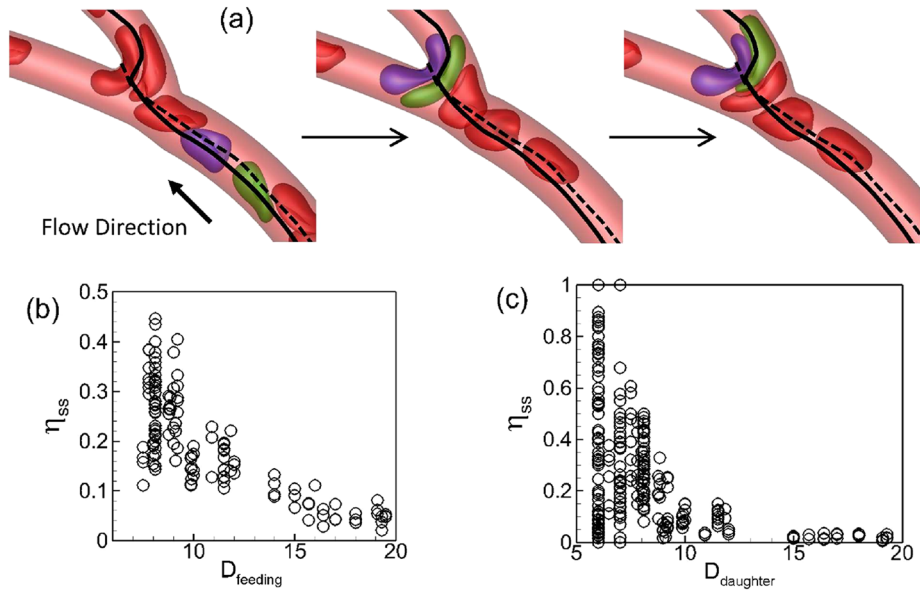


FIG. 15. (a) An example of the “separation-surface-crossover” component of the cell screening mechanism. A time sequence of events is shown. The RBC shown in purple straddles around the bifurcation partially blocking the left daughter branch and forcing the trailing RBC (shown in green) to cross the separation surface and enter the right branch. The dashed line is the separation surface, and the solid line is the trajectory of the RBC in green. [(b) and (c)] The fraction of RBCs (η_{ss}) at the bifurcations that enter the daughter vessels by crossing the separation surface plotted as a function of feeding vessel diameter and daughter vessel diameter, respectively. Diameters are given in μm .

would be almost devoid of cells. Therefore, the separation surface crossover acts as a balancing mechanism and mitigates extreme heterogeneity in hematocrit variation across a vascular network.

These results stem from the ability of cells to deviate from the underlying fluid streamlines. At negligible inertia, cell deformability, as well as cell-cell and cell-vasculature interactions, all contribute to this. The deviation of cell trajectories arising from cell deformation alone can be best studied by considering the motion of an isolated cell, as considered by such studies as Barber *et al.* (2008) or Doyeux *et al.* (2011). Their studies suggest that the separation-surface crossover by cell deformation alone usually only occurs for trajectories that are already close to the separation surface, and this is also observed in our work. Additionally, we find that cell-cell interactions play a major role in separation surface crossover and that such interactions are dominant within the bifurcations. In the study of Barber *et al.* (2011) with two cells, they mentioned that such cell-cell interactions tend to balance the cell distribution. This is in agreement with our finding that such interactions mitigate extreme heterogeneity in hematocrit.

IV. ADDITIONAL DISCUSSIONS

In the present work, only RBCs have been considered. The co-existence of leukocytes, or WBCs, in the blood stream would likely have a considerable impact on the partitioning behavior as these cells traverse a bifurcation. The physiological basis for excluding WBCs from the model is that they are significantly outnumbered by RBCs under healthy conditions, as mentioned previously. The present work is thus representative of a period of time when WBCs are not present in the section of the microvasculature concerned. While our numerical approach can also model WBCs, as was shown in Balogh and Bagchi (2017a), at present, this is a topic of future work.

As to the potential impact of the additional presence of leukocytes on the present study, a few comments can be made. In a more general sense, the time scales associate with the transit time of a WBC through a network such as those considered

here would likely be orders of magnitude higher than that of the RBCs. This is mainly on account of their larger size causing an increased transit time. Thus the partitioning results, in general, would have a different temporal basis than that presented here based on RBCs alone. In a more specific sense, some of the cellular mechanisms could be impacted. Mechanism 2 could occur instead as the result of a WBC partial blockage at an upstream bifurcation in a manner similar to that achieved by the RBCs with this mechanism. This would similarly cause the RBCs to flow into the feeding vessel by moving around the obstruction. The presence of WBCs would also likely enhance the asymmetry in resistance between daughter vessels, which is similar to what occurs with mechanism 3 and would likely be more severe. With mechanism 4, the temporary flow balancing from RBCs piling up at the bifurcation could also occur by a WBC blockage in a symmetric manner, although this would likely occur over a longer period of time.

WBCs would impact the hematocrit profiles in that RBCs would either re-route around these cells, or in the case of capillaries where they block the entire vessel, the RBCs would build up behind the WBC. As the WBC then passed through the bifurcation and entered a daughter branch, during this period, the RBC flow would likely increase into the other daughter branch significantly. WBC presence would also add an additional feature to the cell screening component, where the finite-size effects would significantly alter the partitioning behavior. In the extreme case of the capillaries where a WBC could completely block one daughter vessel, the entire partitioning would be due to the cell screening component.

V. CONCLUSIONS

To the best of our knowledge, the current work is the first DNS study to consider the partitioning of RBC suspensions in simulated microvascular networks with some *in vivo*-like features, such as the presence of a large number of sequential bifurcations and winding vessels with variable cross sections. Our model resolves extreme deformation of individual RBC, cell-cell and cell-vasculature interaction in flowing

suspension, and highly complex geometry of microvasculature. Using the DNS results, a detailed analysis is presented on the partitioning behavior as it develops naturally in successive bifurcations. The major findings from this analysis are as follows:

- (i) We find that while the time-averaged partitioning at a bifurcation could be either the classical or reverse type, the time-dependent partitioning may cycle between these two types over time. The time-averaged partitioning then depends on the duration and frequency of these cycles. Therefore, such time-dependency has influence on the ultimate distribution of the RBCs which is known to be heterogeneous in a microvascular network.
- (ii) We identify four different cellular-scale mechanisms underlying the time-dependent partitioning. This specific analysis was centered on the reverse partitioning. The first two mechanisms are due to an asymmetry in the RBC distribution in the feeding vessel caused by cell partitioning at the bifurcation upstream. The first mechanism occurs when cells in the feeding vessel flow along the side that is closer to the upstream bifurcation. The second mechanism occurs when the cells flow along the other side of the feeding vessel due to an accumulation (or piling-up) of cells at the upstream bifurcation. The third and fourth mechanisms occur in capillary bifurcations as they arise due to cell-cell and cell-vasculature interactions. In the third mechanism, a sudden increase in RBC flux in one daughter vessel leads to an increased flow resistance, thereby diverting more plasma to the other daughter vessel and giving rise to the reverse partitioning. The fourth mechanism is a flow-balancing mechanism, and it provides a means of transition between classical partitioning and reverse partitioning. It occurs when the flow is simultaneously reduced in both daughter vessels due to cells piling up at the bifurcation.
- (iii) Using our DNS data, we show that, on average, a positive skewness in the hematocrit profile in the feeding vessel is associated with the classical partitioning, while a negative skewness is associated with the reverse partitioning.
- (iv) We then present a detailed analysis and insights on two components of disproportionate partitioning, namely, plasma skimming and cell screening. To our knowledge, our work is the first work to dissect these two components for a cell suspension flowing through complex networks. Plasma skimming is simulated by considering the advection of a hematocrit profile through the networks. It is shown that both the classical partitioning and reverse partitioning are present in the plasma skimming component. More importantly, however, we show that the plasma skimming component under-predicts the overall cell partitioning, leaving the cell screening component to make up for the difference. The plasma skimming contribution is found to be dependent on the vessel diameter, which is explained using trends in the hematocrit bluntness. It is further observed that both plasma skimming and cell screening are of similar magnitudes.

- (v) We then analyze the two primary mechanisms giving rise to the cell screening component. The first mechanism, which we termed as the “all-or-none” effect, can be thought of as plasma skimming with finite-sized cells and is observed to over-predict the disproportionality. This is countered by the second mechanism which represents cells crossing over the separation surface. We identify two primary means by which cells cross the separation surface, each of which is associated with the time-dependent RBC dynamics near a bifurcation.
- (vi) The separation surface crossover is quantified in terms of the vessel diameter and is shown to be a dominant and physiologically important mechanism. In some bifurcations, nearly half of the RBCs in the feeding vessels are observed to cross the separation surface. Furthermore, for some daughter vessels, nearly all RBCs are observed to enter by crossing the separation surface. Without the presence of this mechanism, these daughter vessels would not receive any RBCs. Thus, the separation surface crossover is a mechanism which mitigates extreme heterogeneity in RBC distribution across a microvascular network.

SUPPLEMENTARY MATERIAL

See [supplementary material](#) for additional data related to Table I and Fig. 11.

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